



Introduction to Macroeconomics

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Solution: Problem set 3

2025

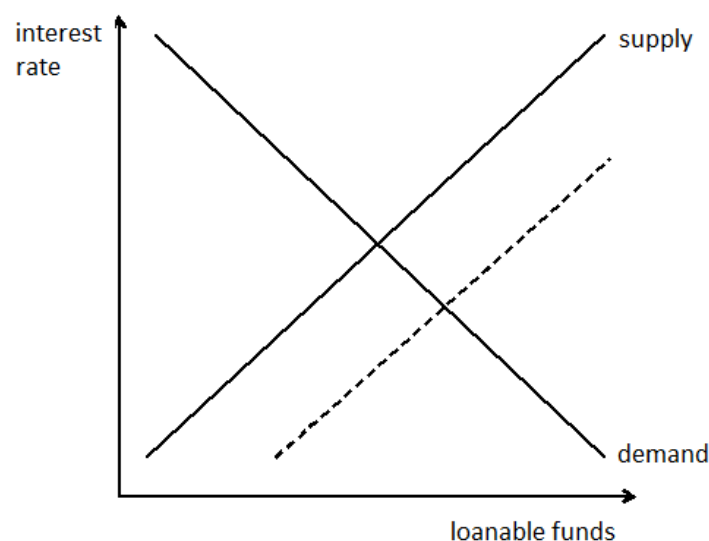
Solution 1

Answer (c)

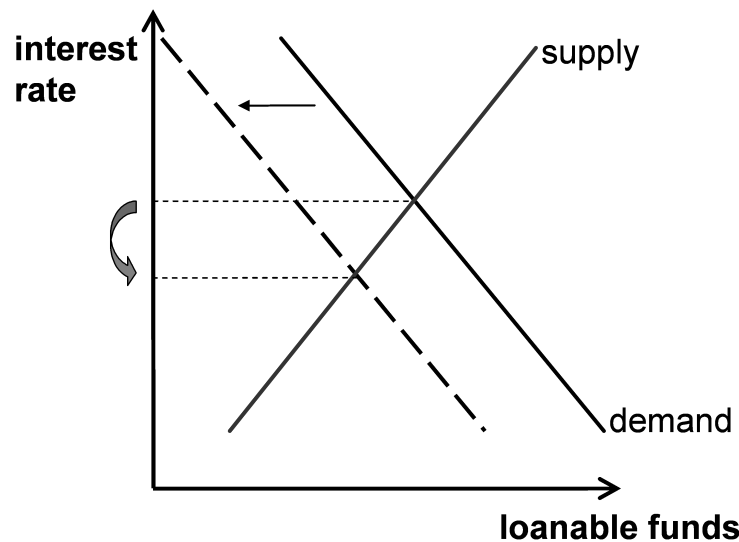
In a closed economy there are no imports or exports and thus GDP is given by : $Y = C + I + G$. Investment equals thus $I = Y - C - G = 400 - 250 - 100 = 50$. Private saving, S_p , is obtained by subtracting consumption and taxes from GDP : $S_p = Y - C - T$. We know that $S_p = 130$, and thus we can calculate taxes : $T = Y - C - S_p = 400 - 250 - 130 = 20$. Public saving equals $S_G = T - G$, i.e. taxes, T , minus public spending, G . We have thus $S_G = T - G = 20 - 100 = -80$, public saving equals -80, which means that the budget deficit equals 80. National saving equals $S = S_p + S_G = 130 + (-80) = 50$. This can be also calculated from the fact that national savings equals investment, $I = S$, and thus $S = 50$.

Solution 2

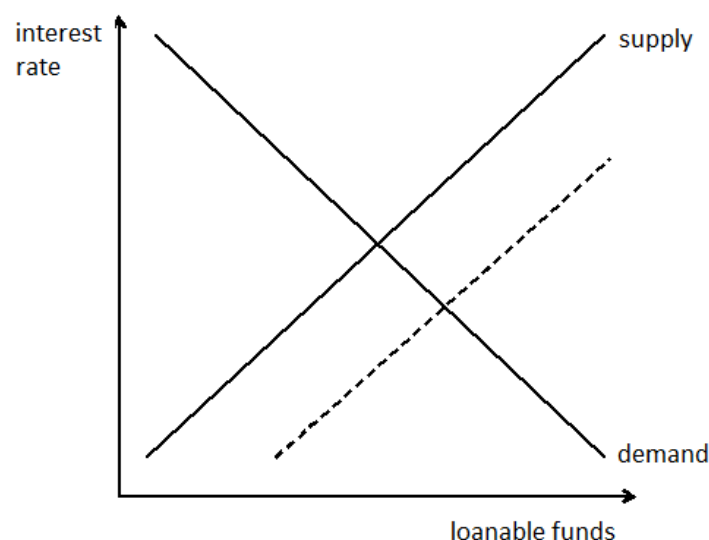
- (a) — If savings by the households increase, the supply of loanable funds would increase for any given level of interest rate. Therefore the supply curve of loanable funds would shift to the right, which would result in a decrease of equilibrium interest rate and an increase in the supply of loanable funds at equilibrium.



- If the investment tax credit disappears, an incentive for firms to invest disappears and, therefore, firms will tend to invest less. As the demand for loanable funds is generated by the firms that wish to invest, the disappearance of the investment tax credit will reduce the demand for loanable funds, and the demand curve will shift to the left. As a result, the equilibrium interest rate falls, and the equilibrium supply of loanable funds decreases.

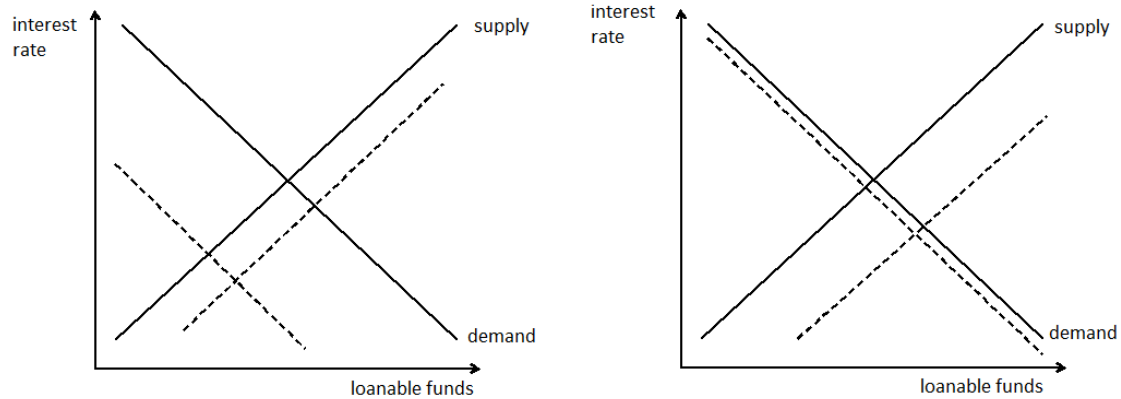


- If a budget surplus appears, it means that the public saving, which was $SG = 0$ (balanced budget) before, now becomes positive. As SG increases, and since total saving is equal to private saving plus public saving ($S = SP + SG$), it must be that total saving increases. The savings generate the supply of loanable funds, and thus we will have a shift of the supply curve to the right, which makes the equilibrium interest rate fall, and the equilibrium supply of loans increase.



- (b) Disappearance of an investment tax credit (decrease in the demand for loanable funds) results in a shift of the demand curve to the left, while the budget surplus (increase in supply of loanable funds) results in a shift of the supply curve to the right. Both of these shifts would necessarily lead to a fall in the equilibrium interest rate. However, the effect of the combination of these two shifts on the equilibrium amount of loans is

unknown. For example, if the shift in demand is larger than the shift in supply, the quantity of loans at equilibrium decreases (as shown in the left graph). On the other hand, if the shift in supply is stronger than the shift in demand, the quantity of loans at equilibrium increases (graph on the right).



Solution 3

When a government decides to borrow to finance its budget deficit, which is generally generated by an increase in the public spending, there is an increase in the demand of loanable funds. As a result, on the market of loanable funds, the interest rate increases. An increase in the interest rate means that loans have become more expensive for firms and, as a result, private investment falls. This phenomenon is referred to as 'crowding out'. Therefore, the 'crowding out' refers to a drop in private investment by firms, not to a drop of the economic activity of private banks. Thus, (a) is false. The fall in private investment does not result from an increase in the taxes on investment, but by the government decision to borrow to finance the budget deficit, which eventually results in an increase in the interest rate. Thus, (b) is false.

Solution 4

The money multiplier is the amount of money the banking system generates with each unit of reserves. Therefore :

$$M_1 = m \times M_0$$

where the money multiplier m is :

$$m = \frac{1}{r + c - rc}$$

However, since r and c are smaller than 1, we can assume that their product, rc , is much smaller than 1, and therefore $rc \ll r + c$. Thus, the multiplier can be simplified to :

$$m = \frac{1}{r + c}$$

A further simplification that is sometimes made is that the households do not hold any cash and deposit all the money in the bank, that is, $c = 0$. The multiplier then becomes :

$$m = \frac{1}{r}$$

We will use these formulas to solve the problem.

a) True :

We need to calculate the money multiplier. We have that r and c are, respectively, 10% and 40%. Thus, we substitute $r = 0, 10$ and $c = 0, 40$ into the expression of the monetary multiplier :

$$m = \frac{1}{r + c} = \frac{1}{0.1 + 0.4} = 2$$

In fact, the money multiplier, without having assumed that $rc \ll r + c$, would have been :

$$m = \frac{1}{r + c - rc} = 2.17$$

Therefore, we can see that the money multiplier is between 2 and 2.2.

b) False

We need to calculate the money supply $M1$ knowing that $M0 = 400\$$. Thus, knowing that $m=2$, $M1$ is :

$$M1 = m \times M0 = 2 \times 400\$ = 800\$$$

And thus, $M1$ does not equal 8000\$ but 800\$.

c) False

If $c \neq 0$ the money multiplier is $m = \frac{1}{r+c}$, while that if $c = 0$ it is $m = \frac{1}{r}$. Since $c > 0$, we can see easily that $\frac{1}{r} > \frac{1}{r+c}$. Therefore, when $c \neq 0$. The value of the multiplier is lower than when $c = 0$.

d) False

Solution 5

Answer : (a)

If the central bank buys bonds, it pays for them with money and thus it increases money supply. On the opposite if it sells bonds it decreases money supply. The refinancing rate is the rate at which the central bank lends money to banks if their reserves are too low. If the refinancing rate is high banks are less willing to borrow and money supply decreases- The reserve ratio determines the share of deposits that banks have to retain in form of reserves. If this rate is high it decreases the money creation potential of the bank which decreases money supply.

- (a) Both buying bonds and decreasing the reserve ratio increases money supply and thus money supply will certainly increase.

- (b) Buying bonds increases money supply, but a higher refinancing rate decreases money supply so the overall effect is uncertain.
- (c) Selling bonds decreases money supply, but a lower refinancing rate increases money supply so the overall effect is uncertain.
- (d) A higher reserve ratio decreases money supply, but a lower refinancing rate increases money supply so the overall effect is uncertain.

Solution 6

The efficient market hypothesis states that all publicly known information about a company is incorporated in the price of this company's stock. This includes also information about future changes in the situation of the company.

- (a) It depends : Suppose that the increases in sales of the other firms were due to a positive shock that affected the entire industry. This means that also the sales of firm A have increased and so did profits. Since this shock was unexpected the price of the stocks of company A increases. Now suppose, that only the sales of the other firms increase, i.e. only the competitors of firm A are affected by the positive shock. This means that the competition in the industry of A has intensified and thus the profit of A decreases unexpectedly. This will decrease the price of A's stock.
- (b) Stays unchanged : Since the change in the fiscal legislation was adopted already last year, at the time when the price of the stock was determined it was known that the profits are going to decrease because of it. According to the efficient market hypothesis it was already incorporated in the price and thus the price does not change.
- (c) Stays unchanged : Again the anticipated profit was already known before the announcement and thus no new information becomes available. The price of firm C's stock stays unchanged.
- (d) Increases : Before the ruling of the court, the profit of the firm was uncertain. If the dispute was resolved to the disadvantage of the firm, it would be lower. After the ruling, this possibility disappears, and the expected profit increases and so does the price of D's stock.

Additional exercises - Not covered in seminar

Solution 7

Solution : (d)

- (a) False : an increase in the reserve requirement means that the commercial banks must hold more reserves and, therefore, can lend less out of each euro that is deposited in the bank accounts. As a result, the banking system can create less money and money supply decreases (please note that the money multiplier decreases when r increases).
- (b) False : if the central bank sells the government bonds it owns, it receives money in payment for those bonds. Hence, money supply decreases.
- (c) False : The refinancing rate is the interest rate at which the central bank is willing to lend to commercial banks on a short-term basis. If the refinancing rate increases, the commercial banks have less incentives to ask the central bank for loans. As a result, the central bank ends up lending less money to the commercial banks, and thus money supply decreases.
- (d) Correct : if the central bank purchases foreign currency it needs to create money to pay for the foreign currency. Therefore, the money supply increases.

Solution 8

Solution : (d)

In a closed economy, the net exports are equal to 0, and therefore $Y = C + I + G$. National saving (or saving) is defined as $S = Y - C - G$. Saving is thus the total income that remains after paying for consumption (C) and government purchases (G). C corresponds to private consumption by households, and G is the 'consumption' of goods and services by the government.

Using $Y = C + I + G$ and $S = Y - C - G$, we can see that $S = I$. All the income that is saved in the economy is used for investment.

Saving can be divided into private saving (S_P) and public saving (S_G). We can see it if we introduce the net taxes T into $S = Y - C - G$:

$$\begin{aligned} S &= Y - C - G \\ S &= Y - C - G - T + T \\ S &= (Y - C - T) + (T - G) \end{aligned} \quad S = S_P + S_G$$

where $S_P = Y - C - T$ means that private households save the part of income that is left after consuming and paying taxes, and $S_G = T - G$ means that the government is able to save if the taxes it collects (T) are higher than what it spends on goods and services (G).

The government can run a budget surplus or a budget deficit, depending on whether it manages to save or not :

- budget surplus : the government manages to save : $T > G$ and thus $S_G > 0$.

- budget deficit : the government spends more than the taxes it is able to collect : $T < G$ and thus $S_G < 0$.

Since $Y = C + I + G$:

$$Y = 300, C = 180, I = 50, T = 30$$

$$G = Y - C - I = 300 - 180 - 50 = 70$$

Since $S = I$, we know that saving is $S = 50$. Finally, private and public saving are :

$$S_P = Y - C - T = 300 - 180 - 30 = 90$$

$$S_G = T - G = 30 - 70 = -40$$

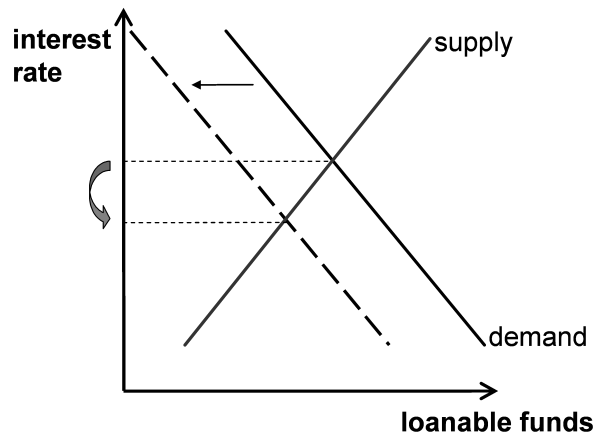
Since $S_G < 0$, the government runs a budget deficit. Let's see if the statements (a) to (d) are correct or not :

- (a) False : $G=70$ and not 30, and $S=50$ and not 60, $S_P = 90$ and not 20, and $S_G = -40$ and not 40.
- (b) False : $G = 70$ and not 30. Since $S_G = -40 < 0$ the government runs a budget deficit of 40.
- (c) False : private saving is $S_P = 90$ and not 40. We do not have a budget surplus of 10, but a budget deficit of 40.
- (d) Correct : none of the previous statements is correct.

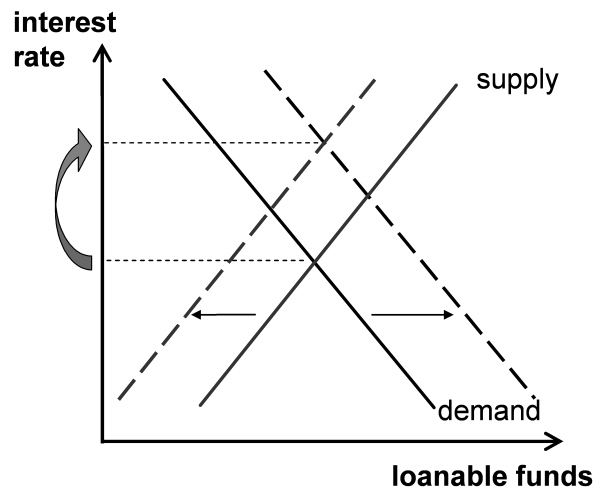
Solution 9

Solution : (b)

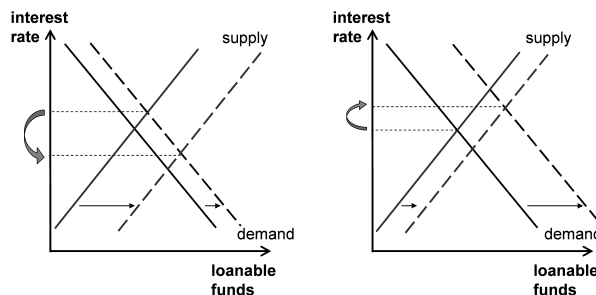
- (a) False : as we can see on the graph, a decrease in the demand for loanable funds implies that the demand curve shifts to the left, and therefore the interest rate decreases.



- (b) Correct : An increase in the demand for loanable funds implies a shift of the demand curve to the right, while a decrease in the supply of loanable funds implies a shift of the supply curve to the left. As we can see on the graph, it leads necessarily to an increase in the interest rate.



- (c) False : a decrease in the demand and in the supply for loanable funds implies that both the demand and the supply curve for loanable funds shift to the left. With the information we have, we cannot know whether the interest rate increases, decreases or remains the same. As we can see on the right-hand graph, if the shift to the left of the demand curve is much more important than the shift to the left of the supply curve then the interest rate rises. However, the left-hand graph shows us that if the shift to the left of the demand curve is much more important than the shift to the left of the supply curve, then the interest rate falls.



- (d) False : an increase in the demand and in the supply for loanable funds implies that both the demand and the supply curve for loanable funds shift to the right. With the information we have, we cannot know whether the interest rate increases, decreases or remains the same. The graphs show, as in (c), that the interest rate can fall or rise depending on the relative importance of the shifts of the curves.